

REPORT PAGES

Executive summary

Prevalence & trends

Who dies

Survival & hazards

Competing risks

Causal & burden

Model performance

Risk & what-if

Data

NHANES 1999-2018 linked to NDI
mortality, follow-up to Dec 2019

Analysis

Python, DuckDB and R (survey,
survival)

Md. Ashiqur Rahman Khan

Executive summary

Cardiometaabolic risk factors and mortality in US adults, NHANES 1999-2018

NHANES 1999-2018 | Follow-up to Dec 2019 | US adults 20+

Participants

50,819

Adults 20+, 10 survey cycles

Deaths

8,279

16.3% of the cohort died

Person-years

480K

Total follow-up time

Median follow-up (years)

9.0

Linked to National Death Index

US adults represented

213.7M

Survey-weighted population

Diabetes prevalence

11.8%

Age-standardized, crude 12.4%

Undiagnosed diabetes

23.4%

Share of all diabetes cases

Diabetes hazard ratio

1.44

95% CI 1.34-1.55

Deaths attributable to CKD

14.7%

Largest attributable share

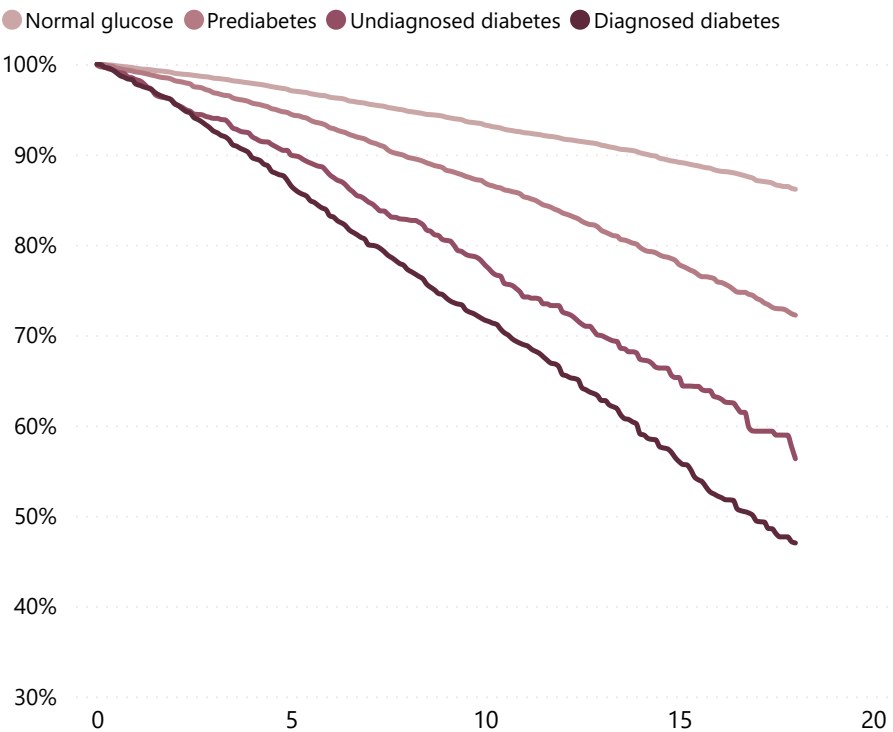
Model C-index

0.872

10-yr risk, temporal validation

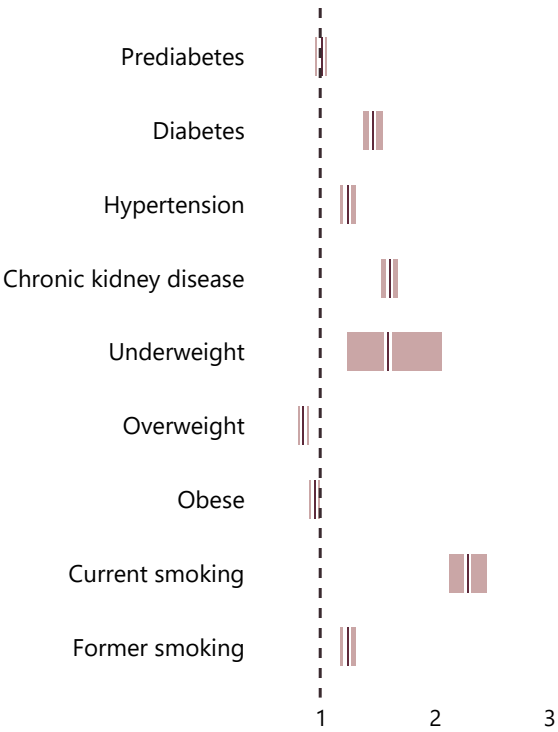
Survival by glycemic status

Kaplan-Meier, survey-weighted, % still alive



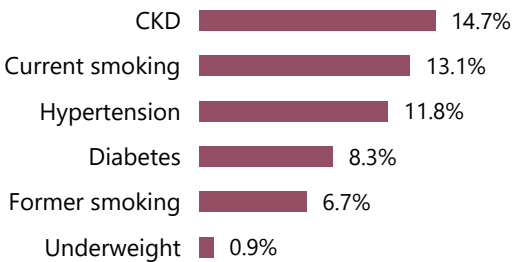
What drives mortality

Hazard ratio with 95% CI, fully adjusted (M3)



Deaths attributable to each factor

Population attributable fraction, all causes



Key findings

1 Diabetes: 44% higher death rate
Prediabetes adds none once age is removed.

2 CKD: 14.7% of all deaths
More than smoking (13.1%) or diabetes (8.3%).

3 Death records miss diabetes
Only 35% with diagnosed diabetes have it on the certificate.

NHANES Mortality

Cardiometabolic risk 1999-2018

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Prevalence & trends

Survey-weighted prevalence of cardiometabolic conditions across 10 NHANES cycles

Estimate

Age-standardized

Breakdown

Race/ethnicity

Reset

Condition

Single select, drives every visual

Pick a condition

Diabetes (total)

SELECTED

Diabetes (total)

GLYCEMIC

By race/ethnicity

Highest

Mexican American 18.6%

Lowest

Non-Hispanic White 9.8%

Highest group is 1.9x the lowest

How to read the estimates

Age-standardized: reweighted to the 2000 US population, so cycles and groups compare fairly.

Crude: the actual share in each cycle, which rises as the population ages.

Tip

Hover any point or bar for the 95% CI. Right-click a bar in All conditions to open that condition's detail page.

Age-standardized prevalence

11.8%

95% CI 11.4-12.2

Crude prevalence

12.4%

For comparison, pooled over 20

Change since 1999-2000

+5.4pts

9.0% to 14.4%, age-standardized

Trend per cycle (odds ratio)

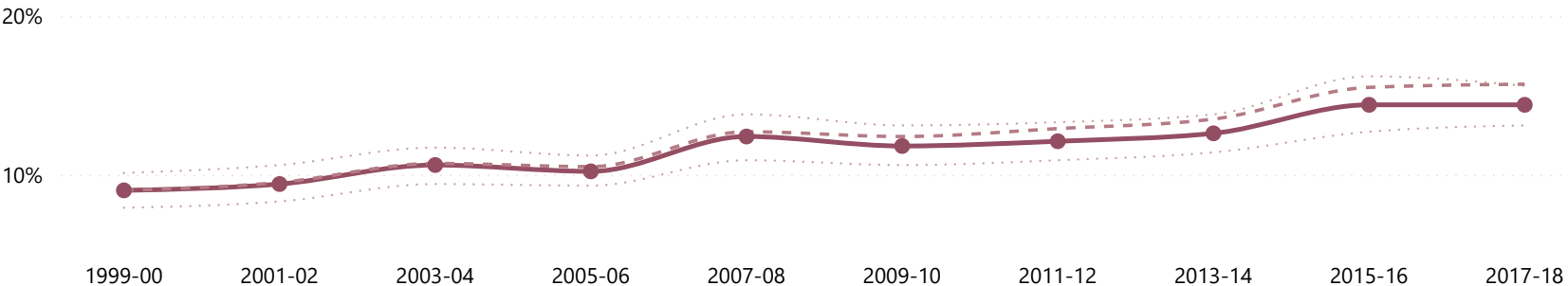
1.056

95% CI 1.041-1.071 | p <0.001

Diabetes (total) prevalence by survey cycle

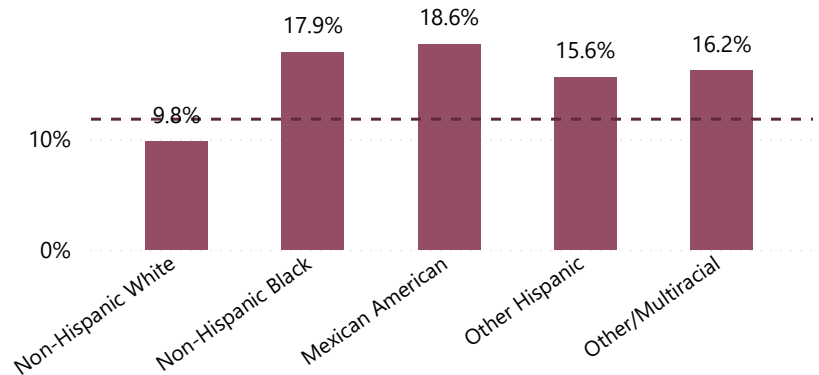
Age-standardized with 95% CI (dotted), crude dashed for comparison

Selected estimate Lower 95% CI Upper 95% CI Other estimate



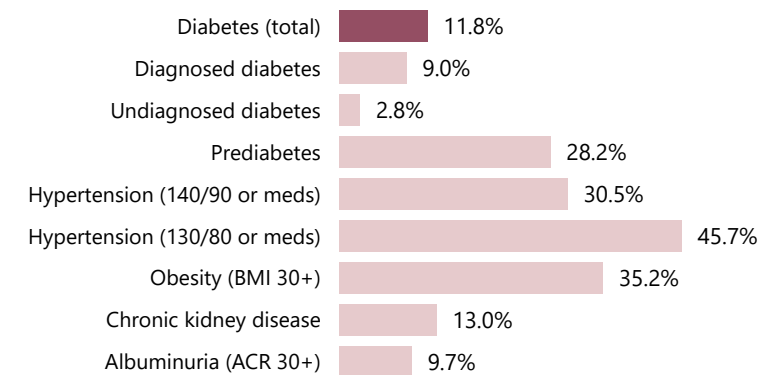
Diabetes (total) by race/ethnicity

Age-standardized %, dashed line = national 11.8%



All conditions at a glance

Age-standardized %, selected condition highlighted



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Who dies: crude vs age-standardized

Crude death rates mix the effect of a condition with the age of the people who have it

Group

All groups

Rate

Both

Reset

Prediabetes looks risky until age is removed

15.1 → 12.1

Per 1,000 PY, close to normal glucose (11.6)

Current smokers look safe until age is removed

12.2 → 18.6

Per 1,000 PY, highest smoking group (never 10.4)

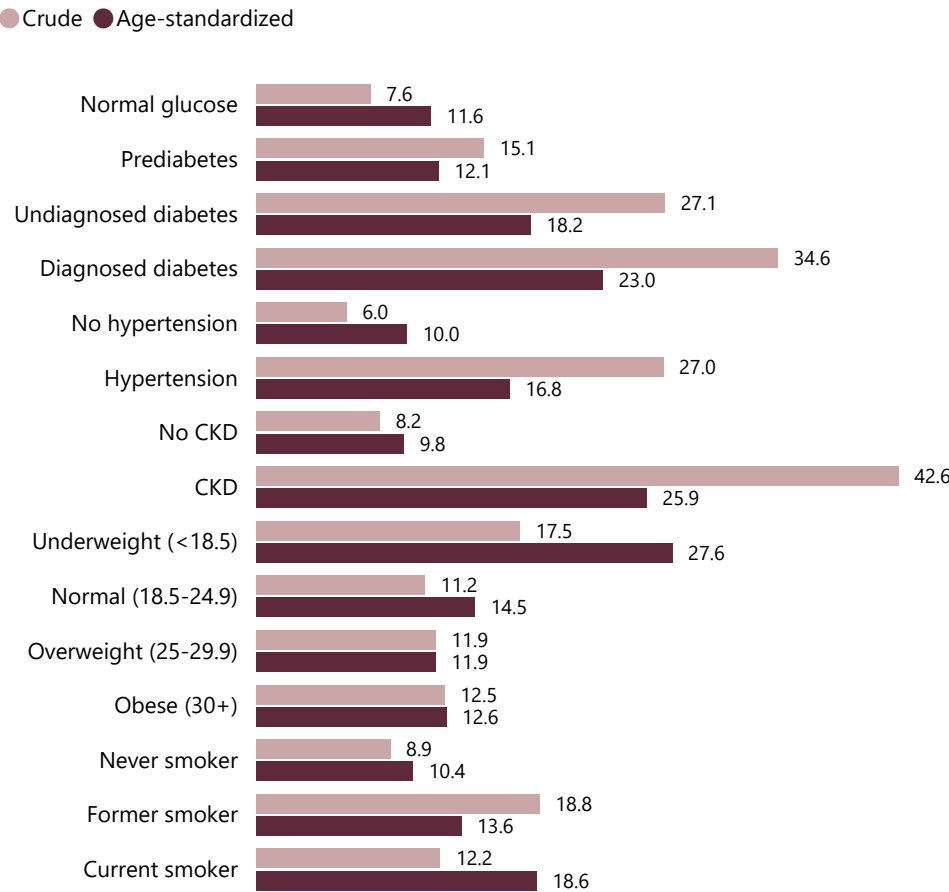
Underweight carries the highest BMI risk

27.6

Per 1,000 PY, twice normal weight (14.5)

Death rate per 1,000 person-years

Crude vs age-standardized to the 2000 US population. Right-click a group to drill through



Baseline characteristics by glycemic status

Survey-weighted mean or %, 95% CI in brackets

Characteristic	Normal	Prediabetes	Diabetes	Overall
Participants, n (unweighted)	26443	15718	8658	50819
Deaths, n (unweighted)	3059	2676	2544	8279
Age, years	42.2 (41.9-42.6)	52.5 (52.1-53.0)	59.2 (58.8-59.7)	47.3 (47.0-47.7)
Sex: female	53.8%	47.1%	48.6%	51.2%
BMI, kg/m2	27.3 (27.2-27.4)	30.0 (29.8-30.1)	32.9 (32.6-33.1)	28.8 (28.6-28.9)
HbA1c, %	5.2 (5.2-5.2)	5.7 (5.6-5.7)	7.2 (7.2-7.3)	5.6 (5.6-5.6)
Hypertension	20.7%	40.5%	65.5%	31.9%
Chronic kidney disease	8.5%	14.5%	35.6%	13.6%

Survival at 5, 10 and 15 years

% still alive, survey-weighted Kaplan-Meier

Group	5 years	10 years	15 years
Normal glucose	97.0%	93.2%	89.1%
Prediabetes	94.4%	86.8%	77.8%
Undiagnosed diabetes	89.9%	77.7%	65.3%
Diagnosed diabetes	86.5%	71.6%	56.0%
Hypertension	89.8%	77.2%	64.7%
CKD	82.6%	64.4%	50.7%

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Survival & hazard ratios

Survey-weighted Cox models pooled over 20 imputations, with nonlinear dose-response curves

Curve group

Glycemic status

Adjustment

M3

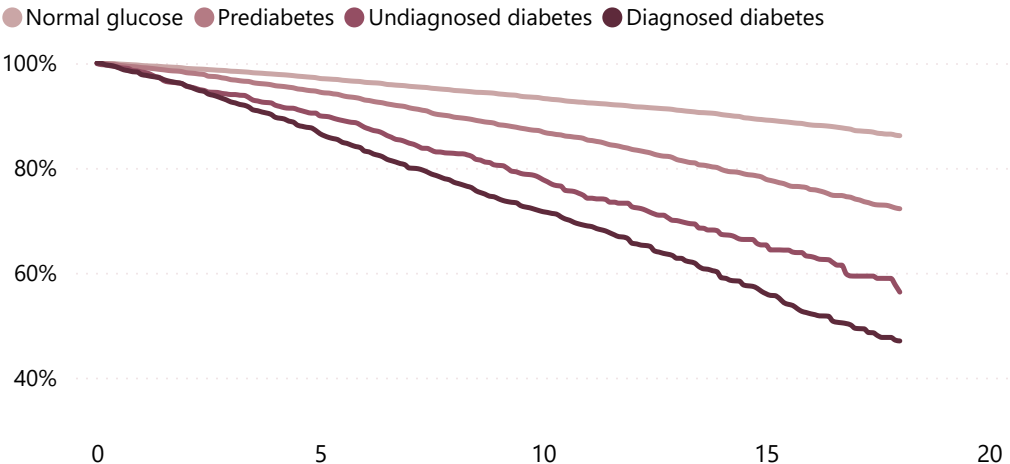
Spline variable

HbA1c (%)

Reset

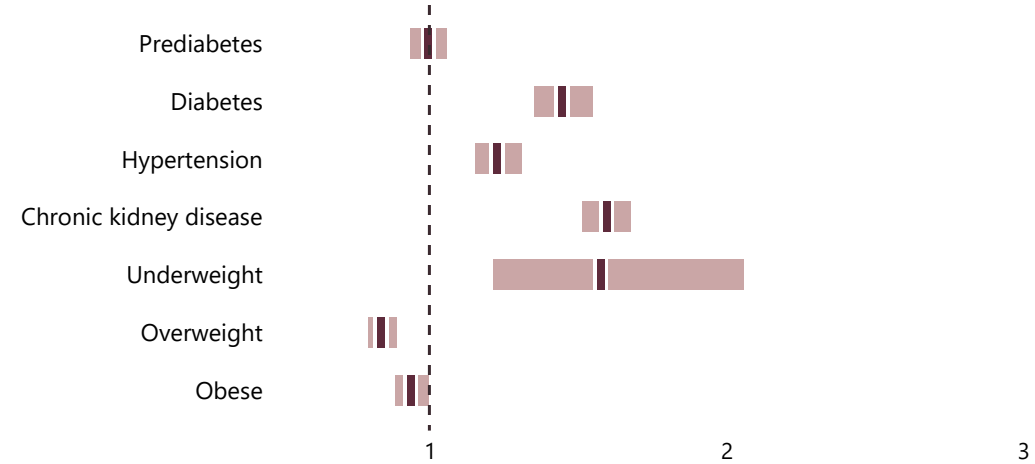
Survival curves by glycemic status

Kaplan-Meier, % still alive over 20 years



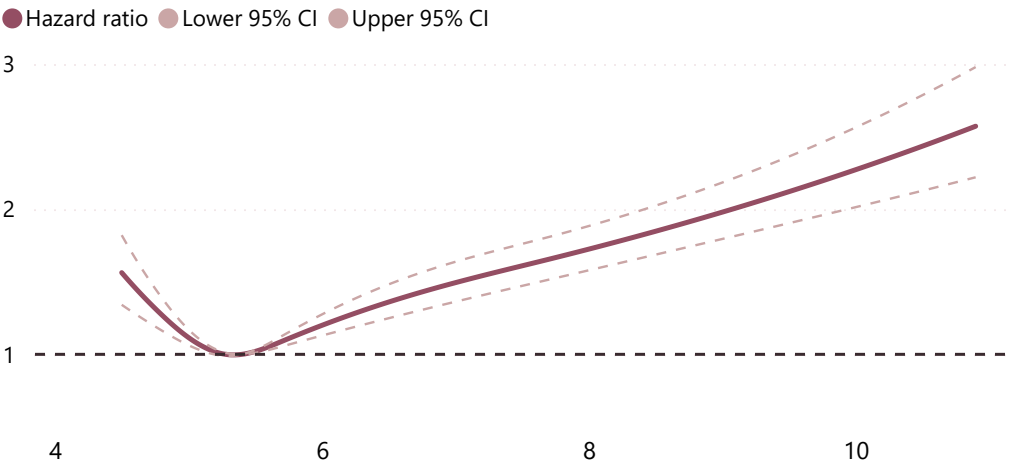
Hazard ratios, model M3

M3: + all conditions



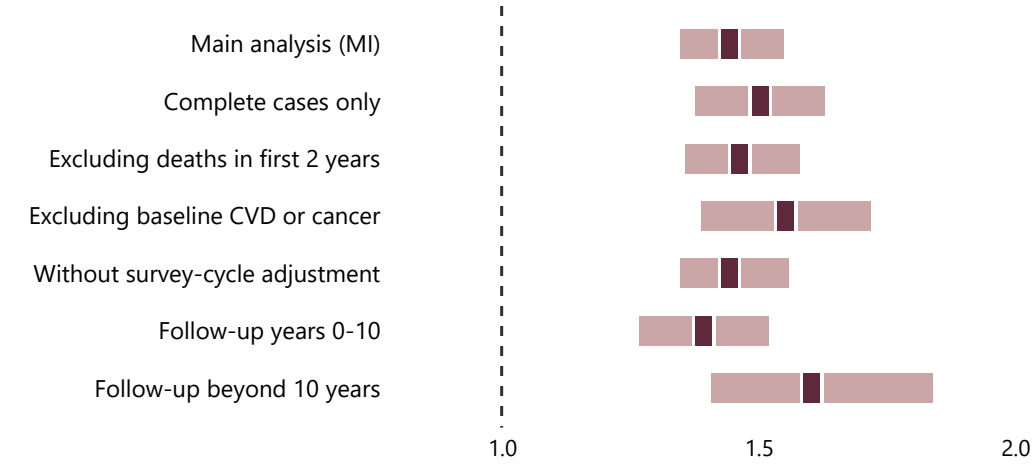
HbA1c (%) and death risk

Restricted cubic spline, dashed lines = 95% CI, dashed rule = HR 1



Is the diabetes result robust?

Diabetes HR (95% CI) across 7 sensitivity analyses



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Competing risks & causes of death
Cause-specific and Fine-Gray models, Aalen-Johansen cumulative incidence

Cause of death

CVD

Group

Glycemic status

HR type

Cause-specific

Reset

Share of deaths, diagnosed DM

35.4%

Selected cause of death

Diabetes HR, selected cause

1.50

95% CI 1.33-1.69, cause-specific

Naive 1-KM overstates 15-yr risk

+19.3%

Diagnosed diabetes, selected cause

Diabetes on the death certificate

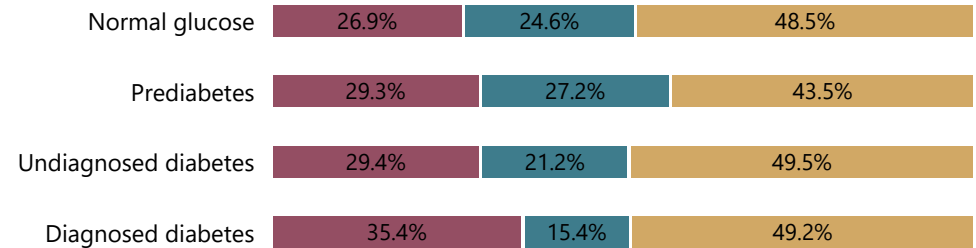
34.5%

Decedents with diagnosed diabetes

Cause of death mix

% of deaths by underlying cause, hover for detail

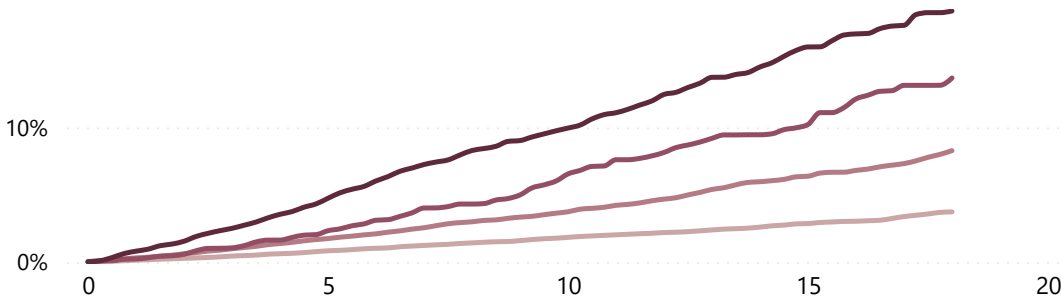
CVD Cancer Other causes



Cumulative incidence of CVD death

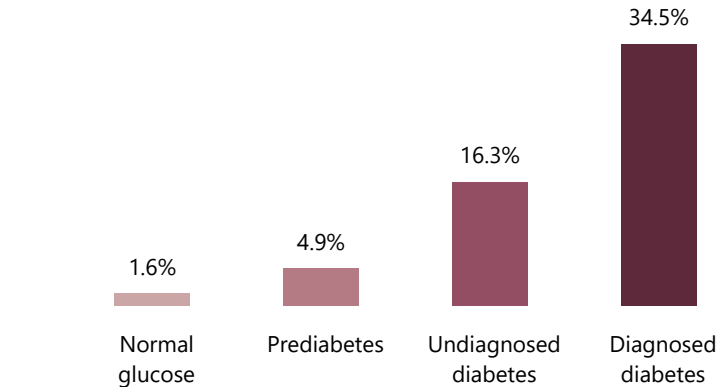
Aalen-Johansen estimate that accounts for competing deaths

Normal glucose Prediabetes Undiagnosed diabetes Diagnosed diabetes



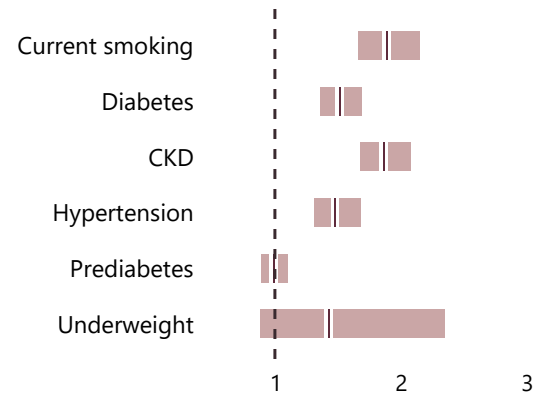
Diabetes on the death certificate

% of decedents with diabetes listed anywhere



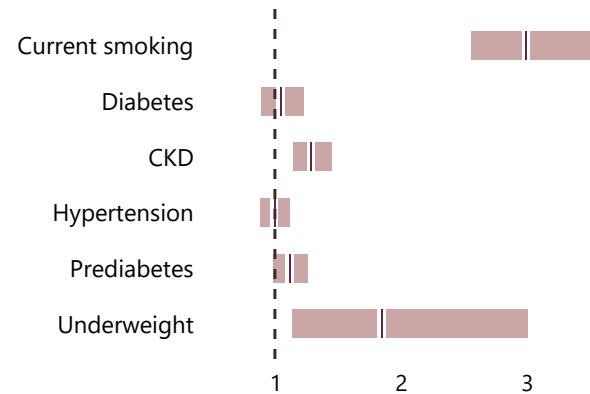
CVD death: hazard ratios

Cause-specific hazard ratios



Cancer death: hazard ratios

Cause-specific hazard ratios



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Causal effects & attributable burden

IPW, E-values for unmeasured confounding and population attributable fractions

Exposure

Diabetes

PAF cause

All causes

Reset

IPW marginal hazard ratio

1.83

95% CI 1.58-2.13

10-year risk difference

+6.3pts

9.9% to 16.2%

E-value

2.41

Strength needed to explain away

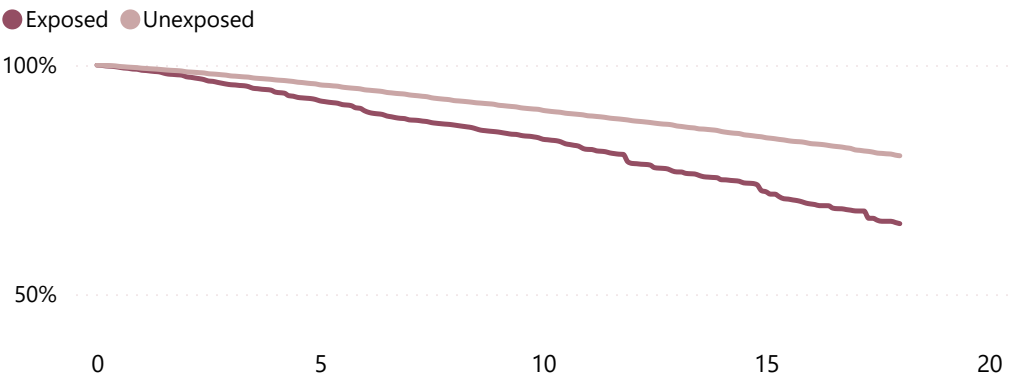
Deaths attributable to exposure

8.3%

Selected exposure and cause

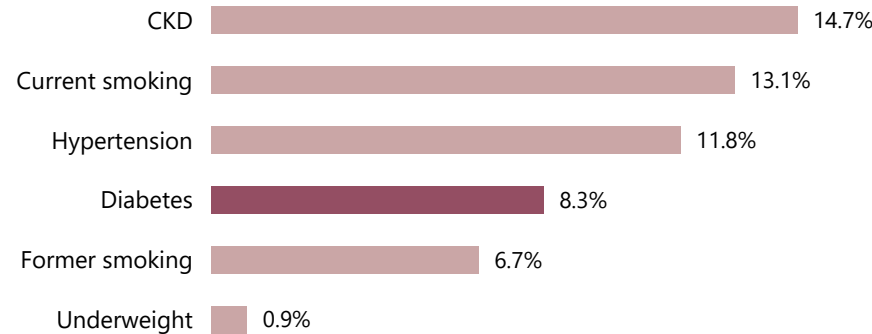
Adjusted survival, diabetes vs none

IPW-weighted survival curves, stabilized weights



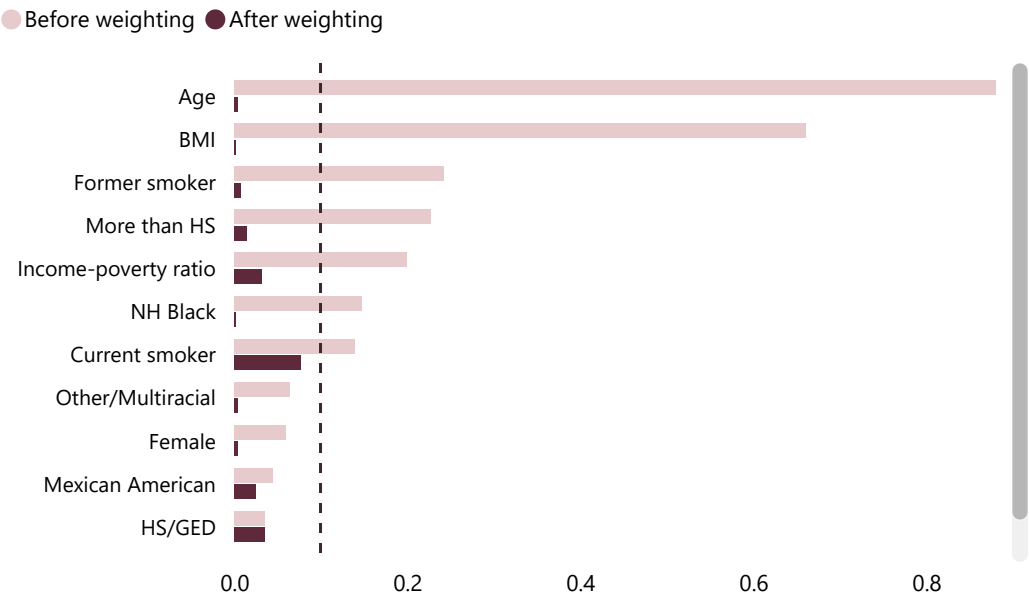
Population attributable fraction

% of deaths that would not occur without the factor, exposure highlighted

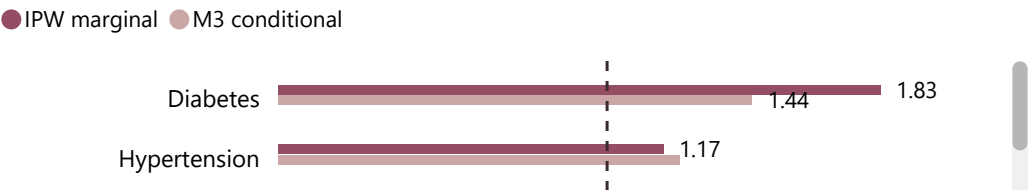


Covariate balance before and after weighting

Absolute standardized mean difference, dashed rule = 0.1 target



Marginal (IPW) vs conditional (M3) hazard ratio



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Model performance

10-year death risk models, developed on 1999-2006 and validated on 2007-2010

Prediction model (multi-select)

All

Reset

Model A: Basic

Age, sex, race/ethnicity, smoking

C-index	Brier	O/E	Slope
0.850	0.0678	1.003	0.988

Model B: Conditions

Basic + glycemic, hypertension, CKD, BMI, prior CVD

C-index	Brier	O/E	Slope
0.862	0.0653	0.995	0.951

Model C: Full

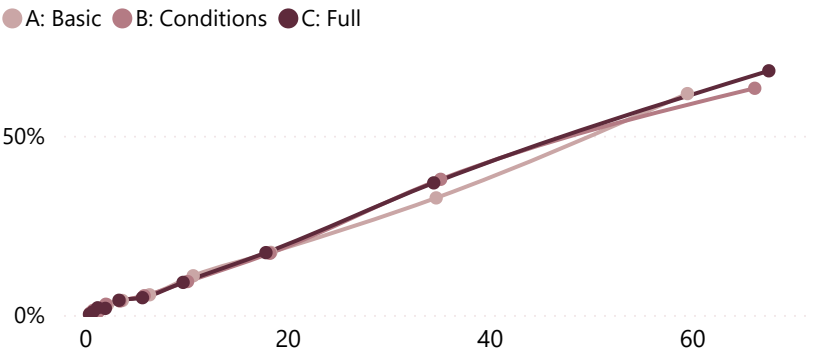
Basic + education, income, biomarkers, treatment

C-index	Brier	O/E	Slope
0.872	0.0622	1.014	0.983

Best

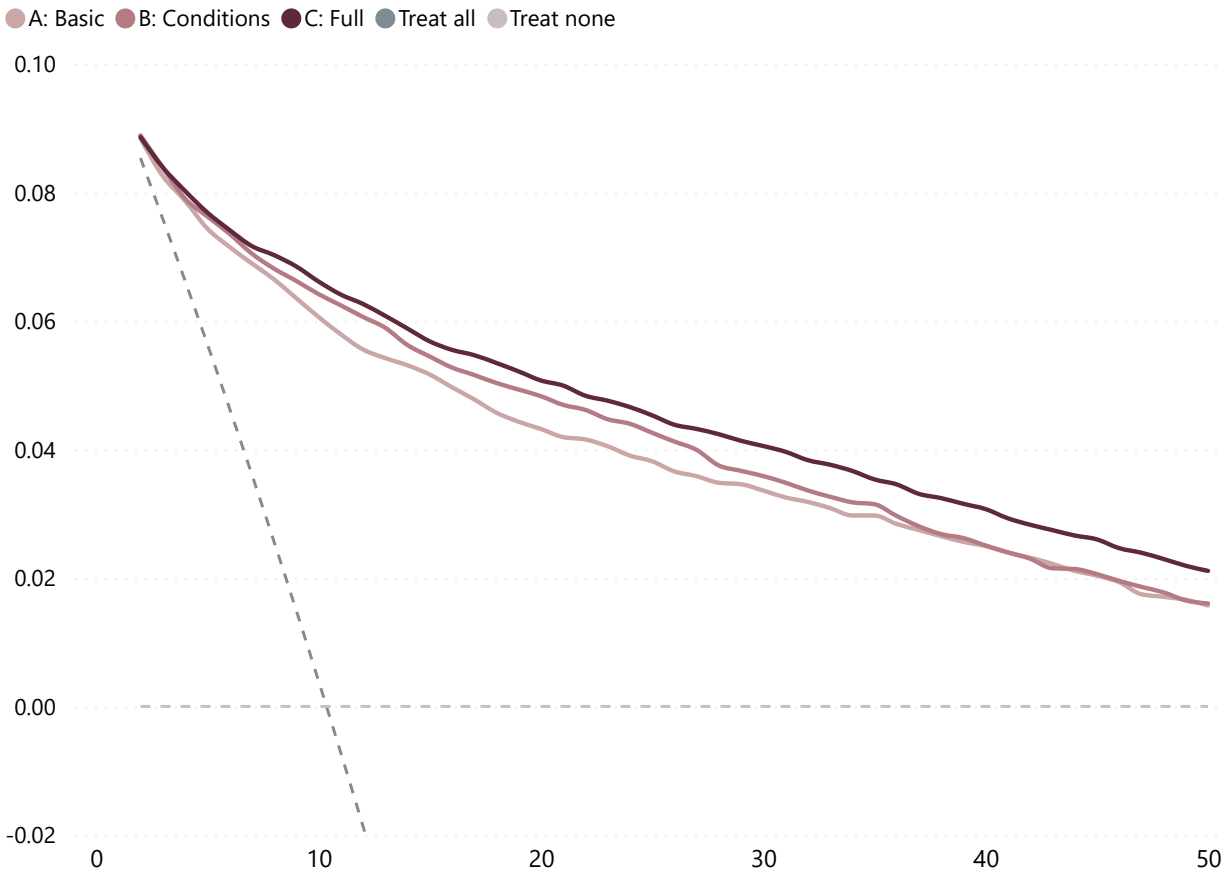
Calibration by risk decile

Observed vs predicted 10-year death %



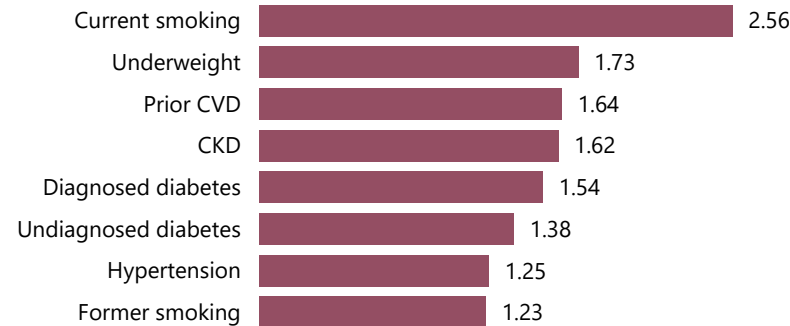
Decision curve analysis

Net benefit across risk thresholds, compared with treat all and treat none



Model B hazard ratios

Top predictors, exponentiated coefficients



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Risk tiers & what-if

Build a profile to see the predicted 10-year risk of death and what would lower it

Model B: Conditions | Prior CVD and cancer fixed at No

Reset...

Build a profile

Pick one value in every slicer

Age

60 years

Sex

Male

Race/ethnicity

Non-Hispanic White

Smoking

Current

Glycemic status

Diagnosed diabetes

Hypertension

Yes

Chronic kidney disease

No

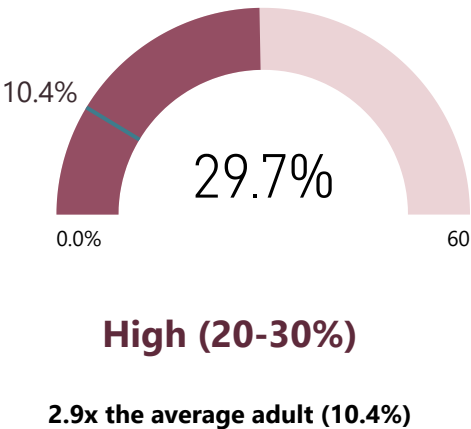
BMI

Obese

How it works
Risks come from a grid of 11,520 profiles scored in R. Nothing is recalculated in DAX.

Predicted 10-year risk of death

Teal mark = average adult (10.4%)



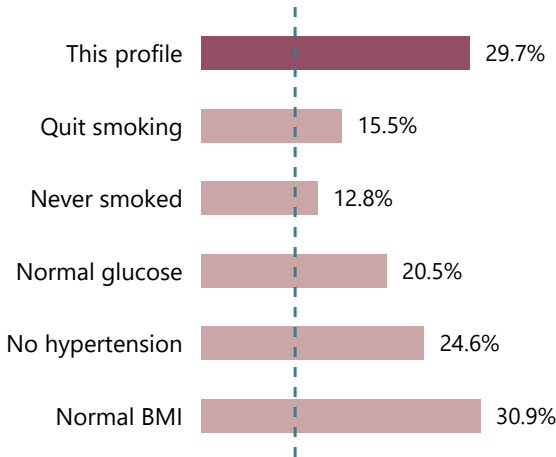
Risk tiers in the validation cohort

Right-click a tier to drill through to its profile. Pred. = predicted, Obs. = observed 10-yr deaths

Risk tier	Range	Pop.	Age	Pred.	Obs.	DM	HTN	CKD
Low	<5%	61.4%	36.9	1.8%	1.7%	4.3%	14.3%	4.8%
Borderline	5-10%	13.3%	55.1	7.1%	6.9%	16.3%	44.7%	10.7%
Intermediate	10-20%	10.0%	61.5	14.2%	13.6%	23.7%	57.6%	19.0%
High	20-30%	4.4%	67.0	24.6%	28.0%	30.8%	67.4%	28.8%
Very high	30%+	11.0%	75.1	56.1%	58.7%	35.3%	77.7%	55.4%

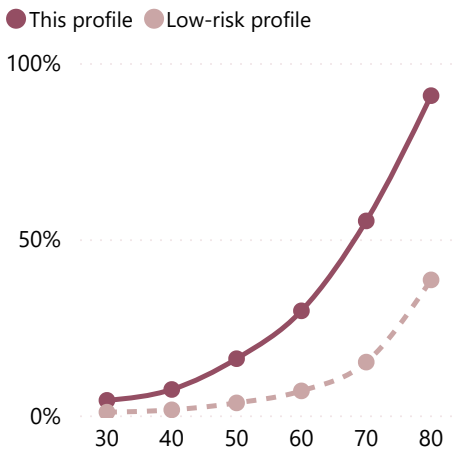
What would lower it

Same profile with one factor changed



Risk by age

Vs a low-risk profile



Predicted vs observed by tier

10-year death %. Right-click a tier to drill through

